

Week 14: Transformers, LLMs, Foundation Models

Welcome to the **Transformers, LLMs, and Foundation Models** module!

This is the point where everything you have built so far, i.e., RNNs, LSTMs, Seq2Seq models, and attention, converges into the architecture that powers modern AI: **the Transformer**.

This module is structured as an **8-day sequential roadmap**, moving from the mechanics of attention through the full Transformer architecture, then to LLMs, their evaluation, and finally to practical fine-tuning

 **Important:** Please follow the days in the *exact order* provided in the **Online Learning Roadmap** to fully grasp the theory and hands-on labs.

By the end of this module, you will have moved from reading the original "**Attention Is All You Need**" paper to **fine-tuning** a real transformer family of models on a production-style routing task. Find the reference Stanford CME295 lecture slides at cme295.stanford.edu/syllabus.

Learning Objectives

- **Explain** why attention was invented and how scaled dot-product attention solves the Seq2Seq bottleneck
- **Describe** every component of the Transformer, i.e., multi-head attention, positional encoding, and encoder-decoder stack.
- **Distinguish** encoder-only (BERT), decoder-only (GPT), and encoder-decoder (T5) architectures and their use cases
- **Articulate** how LLMs are trained, scaled, tuned, and evaluated.
- **Implement** attention in PyTorch from scratch and fine-tune Encoder-Only Models like DistilBERT for a classification task.
- **Fine-tune** an LLM for text generation/classification using the HuggingFace ecosystem.
- **Compare** prompting, BERT fine-tuning, and LLM fine-tuning on accuracy, latency, and cost for a routing task.

Online Learning Roadmap

- **Day 1: NLP Overview | Why Attention? Transformer Concepts**
 - [Jay Alammar — Visualizing NMT: Mechanics of Seq2Seq Models With Attention](#) (article, ~20 min)
 - **[MUST WATCH]** [Stanford CME295 — Lecture 1: Transformer](#) (video, ~60–70 min)
 - [\[slides\]](#) Lecture 1: Transformer
 - [Jay Alammar — The Illustrated Transformer](#) (article, ~25 min)
 - **[Optional]** [Professor Bryce — Transformer and Self-Attention \(DL-19\)](#) (video, ~18 min)
- **Day 2: The Transformer Architecture (Core Theory).**
 - **[MUST WATCH]** [Stanford CME295 — Lecture 2: Transformer-Based Models & Tricks](#) (video, 1h47m)
 - [\[slides\]](#) Lecture 2: Transformer-Based Models & Tricks
 - **[Paper]** [Attention Is All You Need](#)
 - **[3Blue 1Brown Visualization] Attention & Transformers:**
 - [3Blue1Brown — Transformers, the tech behind LLMs \(Ch. 5\)](#) (video, ~27 min)
 - [3Blue1Brown — Attention in transformers, step-by-step \(Ch. 6\)](#) (video, ~26 min)
 - **[Optional]** [Lilian Weng — Attention Recap](#) (article, ~15 min)
 - **[Optional]** [Lilian Weng — The Transformer Family \(technical\)](#)

Note: Watch until 1:02:44, which covers part 1 of Lecture 2, and revisit and complete the remaining content in Day 4

- **Day 3: Hands-On Lab: Attention & Transformers in Code**
 - [DeepLearning.AI — Attention in Transformers: Concepts and Code in PyTorch](#) (short course, 1h16m)
 - [DeepLearning.AI — How Transformer LLMs Work?](#) (short course, 1h44m)

Note: Enroll in free courses and complete the Hands-On lab along with the Instructor guidance.

- **Day 4: Transfer Learning & BERT**

- **[MUST WATCH]** [Stanford CME295 — Lecture 2: Transformer-Based Models & Tricks](#) (video, 1h47m)

- [\[slides\]](#) Lecture 2: Transformer-Based Models & Tricks

***Note:** Continue the remaining content of Lecture 2 that was skipped in Day 2.*

- [Jay Alammar — The Illustrated BERT, ELMo, and Co. \(How NLP Cracked Transfer Learning\)](#) (article, ~25 min)
- [BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding](#) (paper)

- **Day 5: Large Language Models (LLMs)**

- [3Blue1Brown — Large Language Models Explained Briefly](#) (video, ~8 min)

- **[MUST WATCH]** [Stanford CME295 — Lecture 3: Transformers & Large Language Models](#) (video, 1h48m)

- [\[slides\]](#) Lecture 3: Transformers & Large Language Models

- **[MUST WATCH]** [Stanford CME295 — Lecture 4: LLM Training](#) (video, ~1h47m)

- [\[slides\]](#) Lecture 4: LLM Training

- **Day 6: LLM Evaluation | Trending Topics**

- **[MUST WATCH]** [Stanford CME295 — Lecture 8: LLM Evaluation](#) (video, ~1h49m)

- [\[slides\]](#) Lecture 8: LLM Evaluation

- **[MUST WATCH]** [Stanford CME295 — Lecture 9: Recap & Current Trends](#) (video, ~1h51m)

- [\[slides\]](#) Lecture 9: Current trends

- **Day 7: Hands-On Lab: Fine-Tuning | Based on Assignment Tasks**

- **BERT Fine-Tuning:**

- **HuggingFace – FINE-TUNING A PRETRAINED MODEL:**

- [HuggingFace – Processing the data](#)
- [HuggingFace – Fine-Tuning a model with the Trainer API](#)
- [HuggingFace – Write your training loop in PyTorch](#)
- [HuggingFace — Text Classification Guide](#) (hands-on guide, ~20 min — actually run the code)
- [HuggingFace – Understanding and Interpreting Learning Curves](#)

- [Jay Alammar — A Visual Guide to Using BERT for the First Time](#) (article,

~20 min)

- **LLM Fine-Tuning:**
 - [DeepLearning.AI — Fine-Tuning Large Language Models](#) (short course, 1h35m, Beginner-Friendly)
 - [FreeCodeCamp — Fine-Tuning LLM Models](#) (long-form video course, 2h 35m, Intermediate)
 - **Assignment References:**
 - [Fine-Tuning Llama 3.1 for Text Classification | DataCamp](#) (tutorial, ~45 min)
 - [LLM as a Router: How to Fine-Tune Models for Intent-Based Workflows](#) (article, ~15 min — directly relevant to your upcoming assignment)
 - **Hugging Face (LLM Course)**
 - [11. FINE-TUNE LARGE LANGUAGE MODELS](#)
- **Day 8: CME295 | LLM tuning | LLM reasoning | Agentic LLMs**
 - **[MUST WATCH]** [Stanford CME295 – Lecture 7: Agentic LLMs](#)
 - [\[slides\]](#) Lecture 7: Agentic LLMs
 - **[Optional]** [Stanford CME295 – Lecture 5: LLM tuning](#)
 - **[Optional]** [Stanford CME295 – Lecture 6: LLM reasoning](#)